

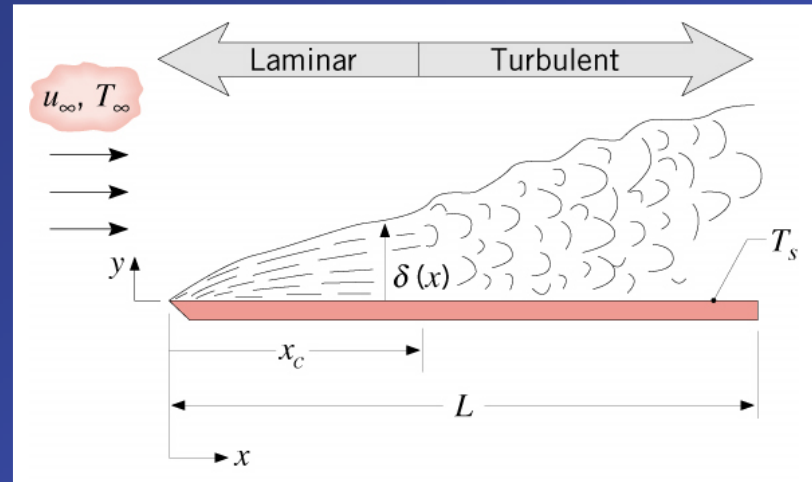
External Flow: The Flat Plate in Parallel Flow

Chapter 7
Section 7.1 through 7.3

TABLE 7.9 Summary of convection heat transfer correlations for external flow^a

Correlation		Geometry	Conditions ^b
$\delta = 5x Re_x^{-1/2}$	(7.17)	Flat plate	Laminar, T_f
$C_{f,x} = 0.664 Re_x^{-1/2}$	(7.18)	Flat plate	Laminar, local, T_f
$Nu_x = 0.332 Re_x^{1/2} Pr^{1/3}$	(7.21)	Flat plate	Laminar, local, T_f , $Pr \geq 0.6$
$\delta_t = \delta Pr^{-1/3}$	(7.22)	Flat plate	Laminar, T_f
$\bar{C}_{f,x} = 1.328 Re_x^{-1/2}$	(7.24)	Flat plate	Laminar, average, T_f
$\bar{Nu}_x = 0.664 Re_x^{1/2} Pr^{1/3}$	(7.25)	Flat plate	Laminar, average, T_f , $Pr \geq 0.6$
$Nu_x = 0.565 Pe_x^{1/2}$	(7.26)	Flat plate	Laminar, local, T_f , $Pr \leq 0.05$, $Pe_x \geq 100$
$C_{f,x} = 0.0592 Re_x^{-1/5}$	(7.28)	Flat plate	Turbulent, local, T_f , $Re_x \leq 10^8$
$\delta = 0.37x Re_x^{-1/5}$	(7.29)	Flat plate	Turbulent, T_f , $Re_x \leq 10^8$
$Nu_x = 0.0296 Re_x^{4/5} Pr^{1/3}$	(7.30)	Flat plate	Turbulent, local, T_f , $Re_x \leq 10^8$, $0.6 \leq Pr \leq 60$
$\bar{C}_{f,L} = 0.074 Re_L^{-1/5} - 1742 Re_L^{-1}$	(7.33)	Flat plate	Mixed, average, T_f , $Re_{x,c} = 5 \times 10^5$, $Re_L \leq 10^8$
$\bar{Nu}_L = (0.037 Re_L^{4/5} - 871) Pr^{1/3}$	(7.31)	Flat plate	Mixed, average, T_f , $Re_{x,c} = 5 \times 10^5$, $Re_L \leq 10^8$, $0.6 \leq Pr \leq 60$
$\bar{Nu}_D = C Re_D^m Pr^{1/3}$ (Table 7.2)	(7.44)	Cylinder	Average, T_f , $0.4 \leq Re_D \leq 4 \times 10^5$, $Pr \geq 0.7$
$\bar{Nu}_D = C Re_D^m Pr^n (Pr/Pr_s)^{1/4}$ (Table 7.4)	(7.45)	Cylinder	Average, T_∞ , $1 \leq Re_D \leq 10^6$, $0.7 \leq Pr \leq 500$
$\bar{Nu}_D = 0.3 + [0.62 Re_D^{1/2} Pr^{1/3} \times [1 + (0.4/Pr)^{2/3}]^{-1/4}] \times [1 + (Re_D/282,000)^{5/8}]^{4/5}$	(7.46)	Cylinder	Average, T_f , $Re_D Pr \geq 0.2$
$\bar{Nu}_D = 2 + (0.4 Re_D^{1/2} + 0.06 Re_D^{2/3}) Pr^{0.4} \times (\mu/\mu_s)^{1/4}$	(7.48)	Sphere	Average, T_∞ , $3.5 \leq Re_D \leq 7.6 \times 10^4$, $0.71 \leq Pr \leq 380$
$\bar{Nu}_D = 2 + 0.6 Re_D^{1/2} Pr^{1/3}$	(7.49)	Falling drop	Average, T_∞
$\bar{Nu}_D = 1.13 C_1 C_2 Re_{D,\max}^m Pr^{1/3}$ (Tables 7.5, 7.6)	(7.52), (7.53)	Tube bank ^c	Average, $\bar{T}_f$, $2000 \leq Re_{D,\max} \leq 4 \times 10^4$, $Pr \geq 0.7$
$\bar{Nu}_D = C C_2 Re_{D,\max}^m Pr^{0.36} (Pr/Pr_s)^{1/4}$ (Tables 7.7, 7.8)	(7.56), (7.57)	Tube bank ^c	Average, $\bar{T}$, $1000 \leq Re_D \leq 2 \times 10^6$, $0.7 \leq Pr \leq 500$

Physical Features



- As with all **external flows**, the boundary layers develop freely without constraint.
- Boundary layer conditions may be entirely laminar, laminar and turbulent, or entirely turbulent.
- To determine the conditions, compute

$$Re_L = \frac{\rho u_\infty L}{\mu} = \frac{u_\infty L}{\nu}$$

and compare with the **critical Reynolds number** for transition to turbulence, $Re_{x,c}$.

$$Re_L < Re_{x,c} \rightarrow \text{laminar flow throughout}$$

$$Re_L > Re_{x,c} \rightarrow \text{transition to turbulent flow at } x_c / L \equiv Re_{x,c} / Re_L$$

- Value of $Re_{x,c}$ depends on free stream turbulence and surface roughness. Nominally,

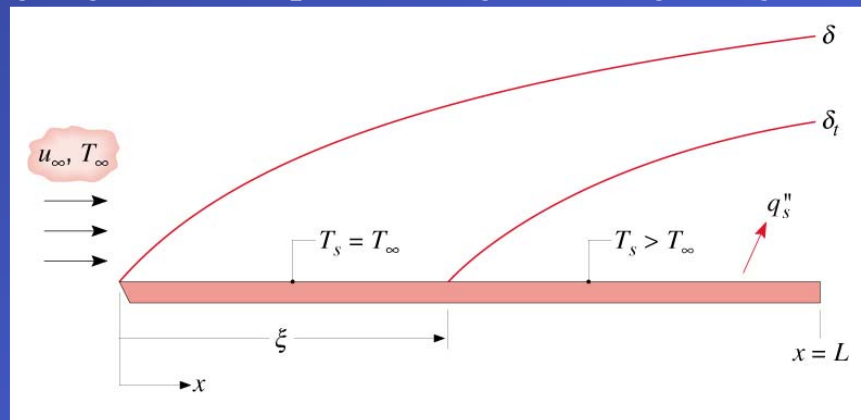
$$Re_{x,c} \approx 5 \times 10^5.$$

- If boundary layer is **tripped** at the leading edge

$$Re_{x,c} = 0$$

and the **flow is turbulent throughout.**

- Surface thermal conditions** are commonly idealized as being of **uniform temperature** T_s or **uniform heat flux** q_s'' . Is it possible for a surface to be concurrently characterized by uniform temperature and uniform heat flux?
- Thermal boundary layer development may be delayed by an **unheated starting length**.



Equivalent surface and free stream temperatures for $x < \xi$ and uniform T_s (or q_s'') for $x > \xi$.

Similarity Solution for Laminar, Constant-Property Flow over an Isothermal Plate

- Based on premise that the dimensionless x -velocity component, u / u_{∞} , and temperature, $T^* \equiv [(T - T_s) / (T_{\infty} - T_s)]$, can be represented exclusively in terms of a **dimensionless similarity parameter**

$$\eta \equiv y(u_{\infty} / \nu x)^{1/2}$$

- Similarity permits transformation of the partial differential equations associated with the transfer of x -momentum and thermal energy to ordinary differential equations of the form

$$2 \frac{d^3 f}{d\eta^3} + f \frac{d^2 f}{d\eta^2} = 0$$

where $(u / u_{\infty}) \equiv df / d\eta$, and

$$\frac{d^2 T^*}{d\eta^2} + \frac{Pr}{2} f \frac{dT^*}{d\eta} = 0$$

- Subject to prescribed boundary conditions, numerical solutions to the momentum and energy equations yield the following results for the **x -component velocity distribution** and the **temperature distribution** in the boundary layer:

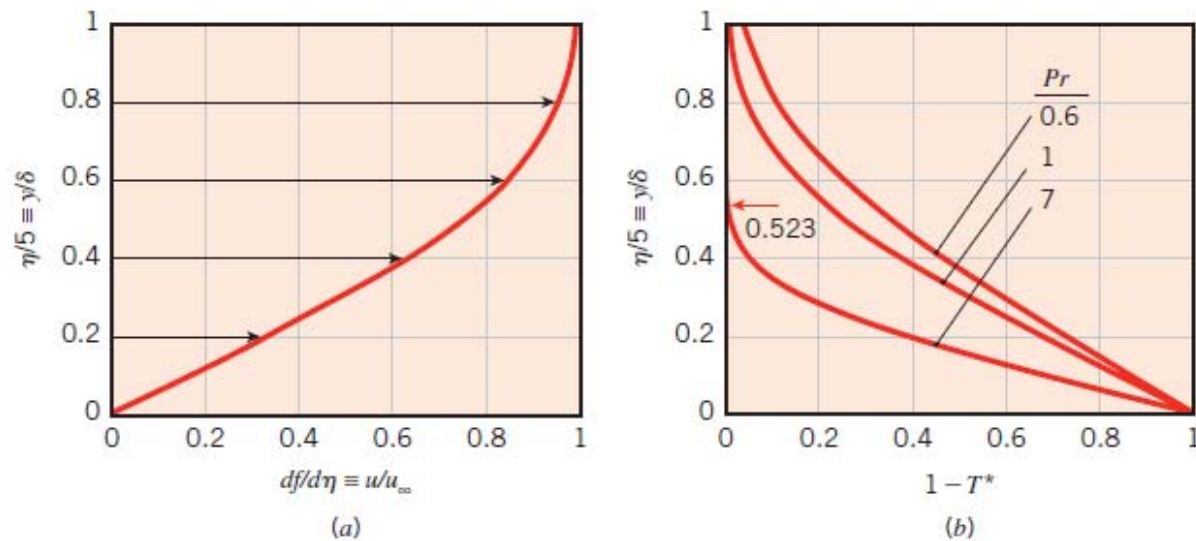


FIGURE 7.4 Similarity solution for laminar flow over an isothermal plate. (a) The x -component of the velocity. (b) Temperature distributions for $Pr = 0.6, 1$, and 7 .

What value of η corresponds to the edge of the velocity boundary layer?

How thick is the thermal boundary layer, relative to the velocity boundary layer, for $Pr = 1$? $Pr > 1$? $Pr < 1$?

With $u / u_\infty = 0.99$ at $\eta = 5.0$ at the edge of the velocity boundary layer,

$$\delta = \frac{5.0}{(u_\infty / \nu x)^{1/2}} = \frac{5x}{(Re_x)^{1/2}}$$

$$\text{With } \tau_s = \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} = \mu u_\infty \sqrt{u_\infty / \nu x} \left. \frac{d^2 f}{d\eta^2} \right|_{\eta=0}$$

$$\text{and } d^2 f / d\eta^2 \Big|_{\eta=0} = 0.332,$$

$$C_{f,x} \equiv \frac{\tau_{s,x}}{\rho u_\infty^2 / 2} = 0.664 Re_x^{-1/2}$$

$$\text{With } h_x = q_s'' / (T_s - T_\infty) = k \partial T^* / \partial y \Big|_{y=0} = k (u_\infty / \nu x)^{1/2} dT^* / d\eta \Big|_{\eta=0}$$

$$\text{and } dT^* / d\eta \Big|_{\eta=0} = 0.332 Pr^{1/3} \text{ for } Pr > 0.6,$$

$$Nu_x = \frac{h_x x}{k} = 0.332 Re_x^{1/2} Pr^{1/3}$$

$$\text{and } \frac{\delta}{\delta_t} = Pr^{1/3}$$

- How would you characterize relative laminar velocity and thermal boundary layer growth for a gas? An oil? A liquid metal?
- How do the local shear stress and convection coefficient vary with distance from the leading edge?
- **Average Boundary Layer Parameters:**

$$\bar{\tau}_{s,x} \equiv \frac{1}{x} \int_0^x \tau_s dx$$

$$\overline{C_{f,x}} = 1.328 Re_x^{-1/2}$$

$$\bar{h}_x = \frac{1}{x} \int_0^x h_x dx$$

$$\overline{Nu_x} = 0.664 Re_x^{1/2} Pr^{1/3}$$

- The effect of variable properties may be considered by evaluating all properties at the **film temperature**.

$$T_f = \frac{T_s + T_\infty}{2}$$

Turbulent Flow

- Local Parameters:

$$\text{Empirical Correlations} \left\{ \begin{array}{l} C_{f,x} = 0.0592 Re_x^{-1/5} \\ Nu_x = 0.0296 Re_x^{4/5} Pr^{1/3} \end{array} \right.$$

How do variations of the local shear stress and convection coefficient with distance from the leading edge for turbulent flow differ from those for laminar flow?

- Average Parameters:

$$\bar{h}_L = \frac{1}{L} \left(\int_0^{x_c} h_{\text{lam}} dx + \int_{x_c}^L h_{\text{turb}} dx \right)$$

Substituting expressions for the local coefficients and **assuming** $Re_{x,c} = 5 \times 10^5$,

$$\bar{C}_{f,L} = \frac{0.074}{Re_L^{1/5}} - \frac{1742}{Re_L}$$

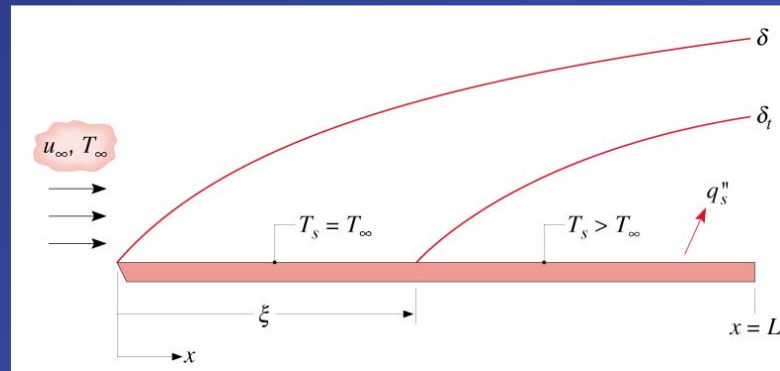
$$\bar{Nu}_L = \left(0.037 Re_L^{4/5} - 871 \right) Pr^{1/3}$$

For $Re_{x,c} = 0$ or $L \gg x_c$ ($Re_L \gg Re_{x,c}$),

$$\bar{C}_{f,L} = 0.074 Re_L^{-1/5}$$

$$\bar{Nu}_L = 0.037 Re_L^{4/5} Pr^{1/3}$$

Special Cases: Unheated Starting Length (USL) and/or Uniform Heat Flux



For both uniform surface temperature (UST) and uniform surface heat flux (USF), the effect of the USL on the **local** Nusselt number may be represented as follows:

		Laminar		Turbulent	
		<u>UST</u>	<u>USF</u>	<u>UST</u>	<u>USF</u>
$Nu_x = \frac{Nu_x _{\xi=0}}{\left[1 - (\xi/x)^a\right]^b}$	a	3/4	3/4	9/10	9/10
	b	1/3	1/3	1/9	1/9
$Nu_x _{\xi=0} = C Re_x^m Pr^{1/3}$	C	0.332	0.453	0.0296	0.0308
	m	1/2	1/2	4/5	4/5

Sketch the variation of h_x versus $(x - \xi)$ for two conditions: $\xi > 0$ and $\xi = 0$.
What effect does an USL have on the local convection coefficient?

- **UST:**

$$q_s'' = h_x (T_s - T_\infty) \quad q = \bar{h}_L A_s (T_s - T_\infty)$$

$$\overline{Nu}_L = \overline{Nu}_L \Big|_{\xi=0} \frac{L}{(L-\xi)} \left[1 - (\xi/L)^{(p+1)/(p+2)} \right]^{p/(p+1)}$$

$p = 2$ for **laminar flow** throughout

$p = 8$ for **turbulent flow** throughout

$\bar{h}_L \rightarrow$ numerical integration for **laminar/turbulent flow**

$$\bar{h}_L = \frac{1}{L} \left[\int_{\xi}^{x_c} h_{\text{lam}} dx + \int_{x_c}^L h_{\text{turb}} dx \right]$$

- **USF:**

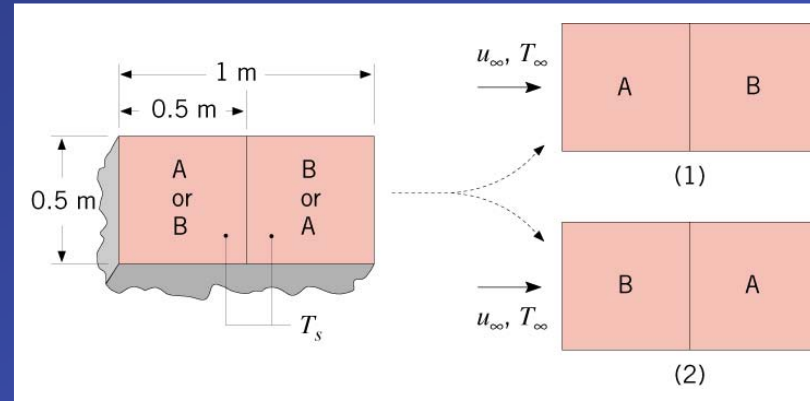
$$T_s = T_\infty + \frac{q_s''}{h_x} \quad q = q_s'' A_s$$

- **Treatment of Non-Constant Property Effects:**

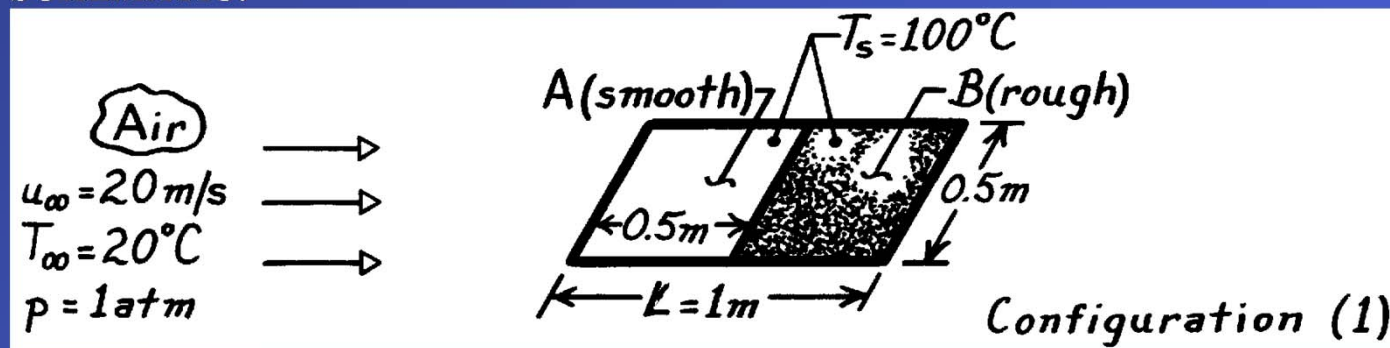
Evaluate properties at the **film temperature**.

$$T_f = \frac{T_s + T_\infty}{2}$$

Problem 7.21: Preferred orientation (corresponding to lower heat loss) and the corresponding heat rate for a surface with adjoining smooth and roughened sections.



SCHEMATIC:



ASSUMPTIONS: (1) Surface B is sufficiently rough to trip the boundary layer when in the upstream position (Configuration 2); (2) $Re_{x,c} \approx 5 \times 10^5$ for flow over A in Configuration 1.

PROPERTIES: *Table A-4*, Air ($T_f = 333\text{K}$, 1 atm): $\nu = 19.2 \times 10^{-6} \text{ m}^2/\text{s}$, $k = 28.7 \times 10^{-3} \text{ W/m}\cdot\text{K}$, $Pr = 0.7$.

ANALYSIS: With

$$Re_L = \frac{u_\infty L}{\nu} = \frac{20 \text{ m/s} \times 1\text{m}}{19.2 \times 10^{-6} \text{ m}^2/\text{s}} = 1.04 \times 10^6$$

transition will occur just before the rough surface ($x_c = 0.48\text{m}$) for Configuration 1. Hence,

$$\overline{Nu}_{L,1} = \left[0.037 \left(1.04 \times 10^6 \right)^{4/5} - 871 \right] 0.7^{1/3} = 1366$$

Since $\overline{Nu}_{L,2} = 0.037 \left(1.04 \times 10^6 \right)^{4/5} (0.7)^{1/3} = 2139 > \overline{Nu}_{L,1}$, it follows that the lowest heat transfer is associated with Configuration 1.

For Configuration 1: $\frac{\overline{h}_{L,1} L}{k} = \overline{Nu}_{L,1} = 1366.$

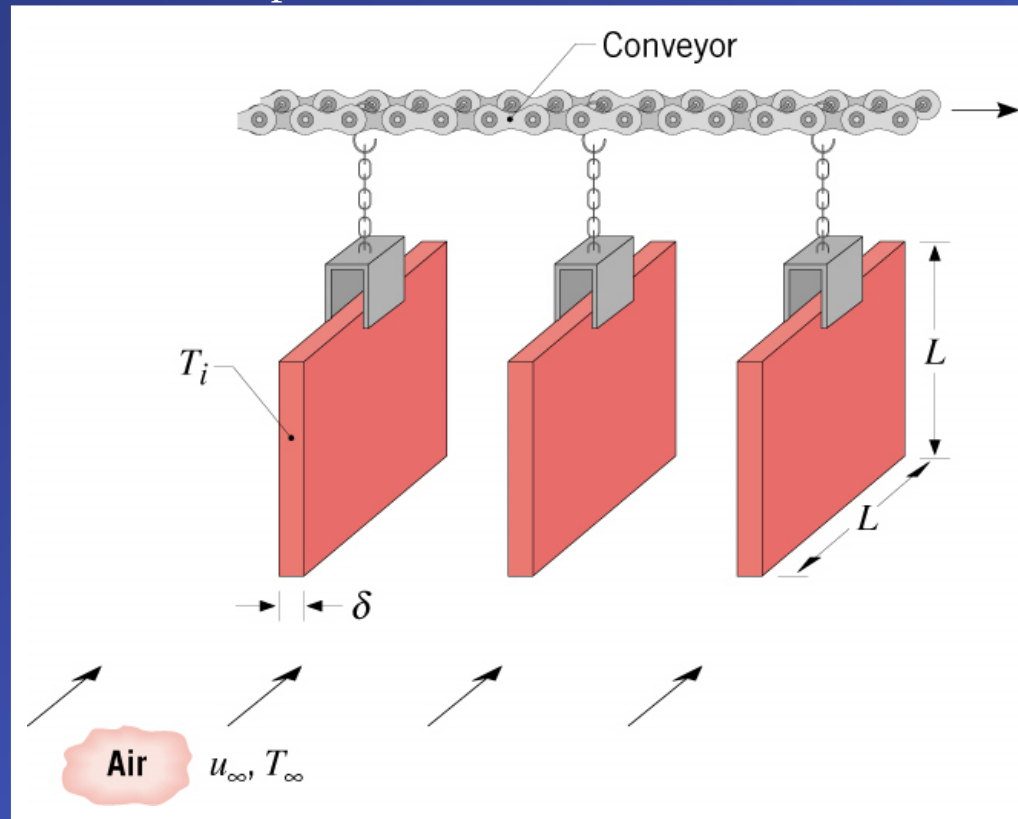
Hence

$$\overline{h}_{L,1} = 1366 \left(28.7 \times 10^{-3} \text{ W/m}\cdot\text{K} \right) / 1\text{m} = 39.2 \text{ W/m}^2 \cdot \text{K}$$

$$q_1 = \overline{h}_{L,1} A (T_s - T_\infty) = 39.2 \text{ W/m}^2 \cdot \text{K} (0.5\text{m} \times 1\text{m}) (100 - 20) \text{K} = 1568 \text{ W} \quad <$$

Comment: For a very short plate, a lower heat loss may be associated with Configuration 2. In fact, parametric calculations reveal that for $L < 30 \text{ mm}$, this configuration provides the preferred orientation.

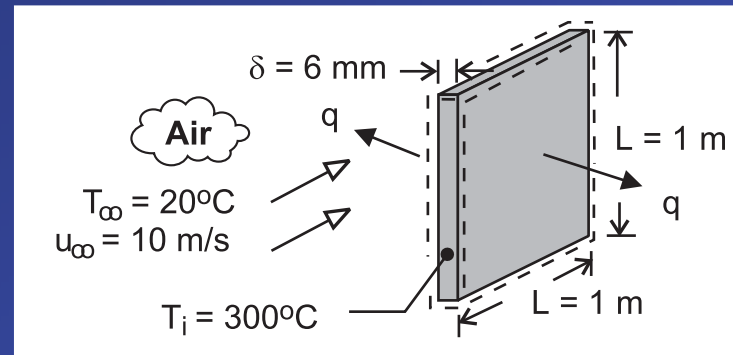
Problem 7.24: Convection cooling of steel plates on a conveyor by air in parallel flow.



KNOWN: Plate dimensions and initial temperature. Velocity and temperature of air in parallel flow over plates.

FIND: Initial rate of heat transfer from plate. Rate of change of plate temperature.

SCHEMATIC:



ASSUMPTIONS: (1) Negligible radiation, (2) Negligible effect of conveyor velocity on boundary layer development, (3) Plates are isothermal, (4) Negligible heat transfer from edges of plate, (5)

$Re_{x,c} = 5 \times 10^5$, (6) Constant properties.

PROPERTIES: Table A-1, AISI 1010 steel (573K): $k_p = 49.2 \text{ W/m}\cdot\text{K}$, $c = 549 \text{ J/kg}\cdot\text{K}$, $\rho = 7832 \text{ kg/m}^3$. Table A-4, Air ($p = 1 \text{ atm}$, $T_f = 433\text{K}$): $\nu = 30.4 \times 10^{-6} \text{ m}^2/\text{s}$, $k = 0.0361 \text{ W/m}\cdot\text{K}$, $Pr = 0.688$.

ANALYSIS: The initial rate of heat transfer from a plate is

$$q = 2 \bar{h} A_s (T_i - T_\infty) = 2 \bar{h} L^2 (T_i - T_\infty)$$

With $Re_L = u_\infty L / \nu = 10 \text{ m/s} \cdot \text{m} / 30.4 \times 10^{-6} \text{ m}^2/\text{s} = 3.29 \times 10^5$, flow is laminar over the entire surface.

Hence,

$$\overline{Nu}_L = 0.664 Re_L^{1/2} Pr^{1/3} = 0.664 (3.29 \times 10^5)^{1/2} (0.688)^{1/3} = 336$$

$$\bar{h} = (k / L) \overline{Nu}_L = (0.0361 \text{ W/m}\cdot\text{K} / 1 \text{ m}) 336 = 12.1 \text{ W/m}^2 \cdot \text{K}$$

$$q = 2 \times 12.1 \text{ W/m}^2 \cdot \text{K} (1 \text{ m})^2 (300 - 20) = 6780 \text{ W}$$

<

Performing an energy balance at an instant of time for a control surface about the plate, $-\dot{E}_{\text{out}} = \dot{E}_{\text{st}}$,

$$\rho \delta L^2 c \left. \frac{dT}{dt} \right|_i = -\bar{h} 2L^2 (T_i - T_\infty)$$

$$\left. \frac{dT}{dt} \right|_i = -\frac{2(12.1 \text{ W/m}^2 \cdot \text{K})(300 - 20) \text{ W/m}^2}{7832 \text{ kg/m}^3 \times 0.006 \text{ m} \times 549 \text{ J/kg} \cdot \text{K}} = -0.26^\circ\text{C/s} \quad <$$

COMMENTS: (1) With $Bi = \bar{h}(\delta/2)/k_p = 7.4 \times 10^{-4}$, use of the lumped capacitance method is appropriate.

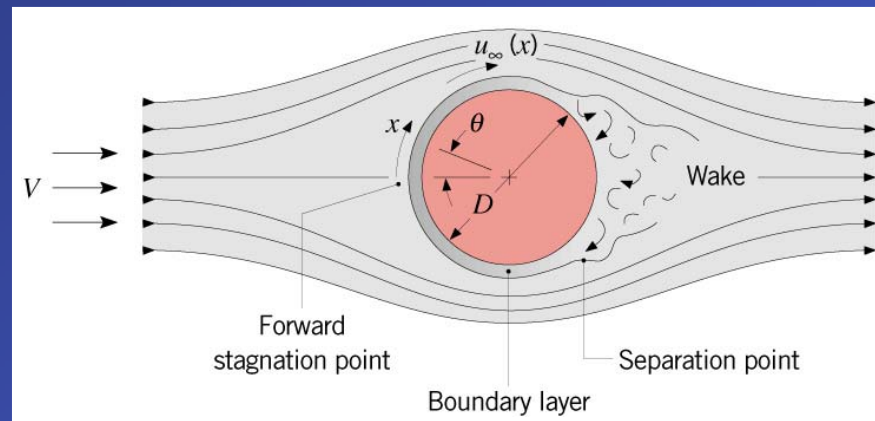
(2) Despite the large plate temperature and the small convection coefficient, if adjoining plates are in close proximity, radiation exchange with the surroundings will be small and the assumption of negligible radiation is justifiable.

External Convection:
Flow over Bluff Objects
(Cylinders, Spheres, Packed Beds)
and
Impinging Jets

Chapter 7
Sections 7.4 through 7.8

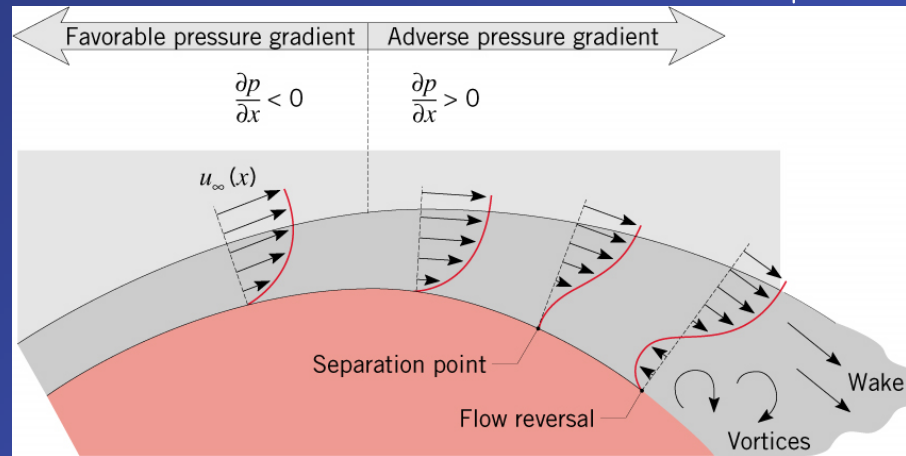
The Cylinder in Cross Flow

- Conditions depend on special features of boundary layer development, including onset at a **stagnation point** and **separation**, as well as **transition** to turbulence.



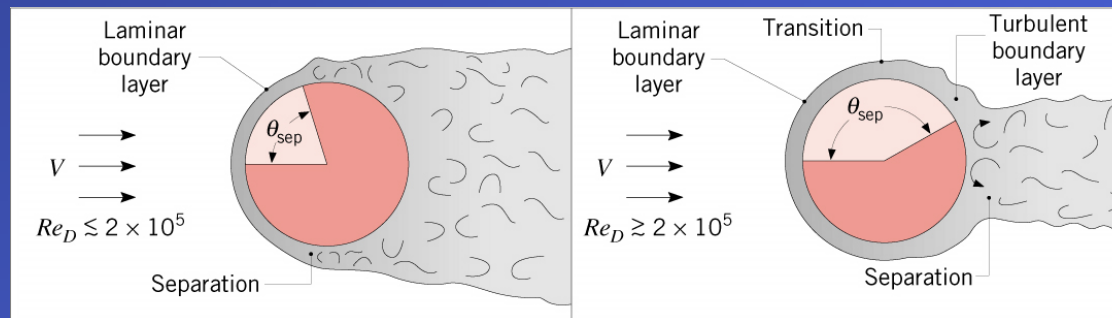
- **Stagnation point**: Location of **zero velocity** ($u_\infty = 0$) and **maximum pressure**.
- Followed by boundary layer development under a **favorable pressure gradient** ($dp/dx < 0$) and hence acceleration of the free stream flow ($du_\infty/dx > 0$).
- As the rear of the cylinder is approached, the pressure must begin to increase. Hence, there is a minimum in the pressure distribution, $p(x)$, after which boundary layer development occurs under the influence of an **adverse pressure gradient** ($dp/dx > 0, du_\infty/dx < 0$).

- **Separation** occurs when the velocity gradient $\left. \frac{du}{dy} \right|_{y=0}$ reduces to zero



and is accompanied by **flow reversal** and a downstream **wake**.

- Location of separation depends on **boundary layer transition**.



$$Re_D \equiv \frac{\rho V D}{\mu} = \frac{V D}{\nu}$$

- What features differentiate boundary development for the flat plate in parallel flow from that for flow over a cylinder?
- Force imposed by the flow is due to the combination of *friction* and *form drag*.
The dimensionless form of the drag force is

$$C_D = \frac{F_D}{A_f \left(\rho V^2 / 2 \right)} \rightarrow \text{Figure 7.9}$$

- Heat Transfer Considerations

- The **Local Nusselt Number**:

- How does the local Nusselt number vary with θ for $Re_D < 2 \times 10^5$?
What conditions are associated with maxima and minima in the variation?
- How does the local Nusselt number vary with θ for $Re_D > 2 \times 10^5$?
What conditions are associated with maxima and minima in the variation?

- The **Average Nusselt Number** ($\overline{Nu}_D \equiv \bar{h}D / k$):

- Churchill and Bernstein Correlation:

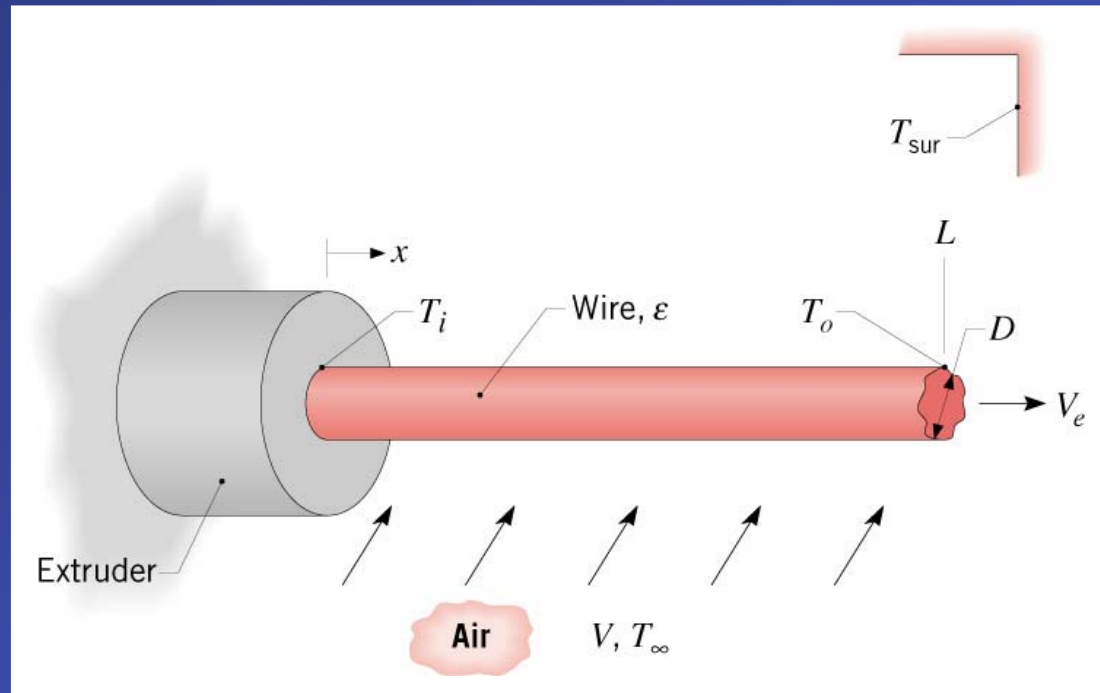
$$\overline{Nu}_D = 0.3 + \frac{0.62 Re_D^{1/2} Pr^{1/3}}{\left[1 + (0.4 / Pr)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{Re_D}{282,000}\right)^{5/8}\right]^{4/5} \quad (7.54)$$

- Cylinders of Noncircular Cross Section:

$$\overline{Nu}_D = C Re_D^m Pr^{1/3}$$

$C, m \rightarrow$ Table 7.3

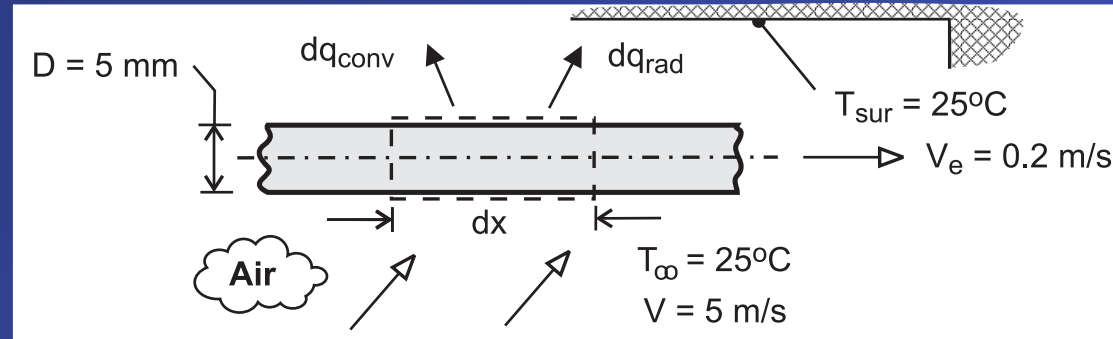
Problem 7.72: Cooling of extruded copper wire by convection and radiation.



KNOWN: Velocity, diameter, initial temperature and properties of extruded wire. Temperature and velocity of air. Temperature of surroundings.

FIND: (a) Differential equation for temperature distribution $T(x)$ along the wire, (b) Exact solution for negligible radiation and corresponding value of temperature at prescribed length ($x = L = 5\text{m}$) of wire, (c) Effect of radiation on temperature of wire at prescribed length. Effect of wire velocity and emissivity on temperature distribution.

SCHEMATIC:



ASSUMPTIONS: (1) Negligible variation of wire temperature in radial direction, (2) Negligible effect of axial conduction along the wire, (3) Constant properties, (4) Radiation exchange between small surface and large enclosure, (5) Motion of wire has a negligible effect on the convection coefficient ($V_e \ll V$).

PROPERTIES: Copper: $\rho = 8900 \text{ kg/m}^3$, $c_p = 400 \text{ J/kg} \cdot \text{K}$, $\varepsilon = 0.55$. Air: $k = 0.037 \text{ W/m} \cdot \text{K}$, $\nu = 3 \times 10^{-5} \text{ m}^2/\text{s}$, $Pr = 0.69$.

ANALYSIS: (a) Applying conservation of energy to a stationary control surface, through which the wire moves, steady-state conditions exist and $\dot{E}_{\text{in}} - \dot{E}_{\text{out}} = 0$.

Hence, with *inflow* due to *advection* and *outflow* due to *advection*, *convection* and *radiation*,

$$\rho V_e A_c c_p T - \rho V_e A_c c_p (T + dT) - dq_{\text{conv}} - dq_{\text{rad}} = 0$$

$$-\rho V_e \left(\pi D^2 / 4 \right) c_p dT - \pi D dx \left[\bar{h} (T - T_\infty) + \varepsilon \sigma (T^4 - T_{\text{sur}}^4) \right] = 0$$

$$\frac{dT}{dx} = -\frac{4}{\rho V_e D c_p} \left[\bar{h} (T - T_\infty) + \varepsilon \sigma (T^4 - T_{\text{sur}}^4) \right] \quad (1) \quad <$$

Alternatively, if the control surface is fixed to the wire, conditions are transient and the energy balance is of the form, $-\dot{E}_{\text{out}} = \dot{E}_{\text{st}}$, or

$$-\pi D dx \left[\bar{h} (T - T_\infty) + \varepsilon \sigma (T^4 - T_{\text{sur}}^4) \right] = \rho \left(\frac{\pi D^2}{4} dx \right) c_p \frac{dT}{dt}$$

$$\frac{dT}{dt} = -\frac{4}{\rho D c_p} \left[\bar{h} (T - T_\infty) + \varepsilon \sigma (T^4 - T_{\text{sur}}^4) \right]$$

Dividing the left- and right-hand sides of the equation by dx/dt and $V_e = dx / dt$, respectively, Eq. (1) is obtained.

(b) Neglecting radiation, separating variables and integrating, Eq. (1) becomes

$$\int_{T_i}^T \frac{dT}{T - T_\infty} = -\frac{4\bar{h}}{\rho V_e D c_p} \int_0^x dx$$

$$\ln \left(\frac{T - T_\infty}{T_i - T_\infty} \right) = -\frac{4\bar{h} x}{\rho V_e D c_p}$$

$$T = T_\infty + (T_i - T_\infty) \exp \left(-\frac{4\bar{h} x}{\rho V_e D c_p} \right) \quad (2) \quad <$$

With $Re_D = VD / \nu = 5 \text{ m/s} \times 0.005 \text{ m} / 3 \times 10^{-5} \text{ m}^2/\text{s} = 833$, the Churchill-Bernstein correlation yields

$$\overline{Nu}_D = 0.3 + \frac{0.62(833)^{1/2} (0.69)^{1/3}}{\left[1 + (0.4/0.69)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{833}{282,000}\right)^{5/8}\right]^{4/5} = 14.4$$

$$\bar{h} = \frac{k}{D} \overline{Nu}_D = \frac{0.037 \text{ W/m} \cdot \text{K}}{0.005 \text{ m}} 14.4 = 107 \text{ W/m}^2 \cdot \text{K}$$

Hence, applying Eq. (2) at $x = L$,

$$T_o = 25^\circ\text{C} + (575^\circ\text{C}) \exp\left(-\frac{4 \times 107 \text{ W/m}^2 \cdot \text{K} \times 5 \text{ m}}{8900 \text{ kg/m}^3 \times 0.2 \text{ m/s} \times 0.005 \text{ m} \times 400 \text{ J/kg} \cdot \text{K}}\right)$$

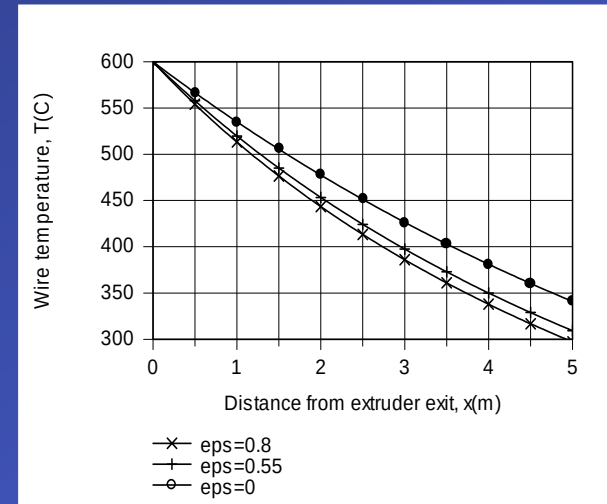
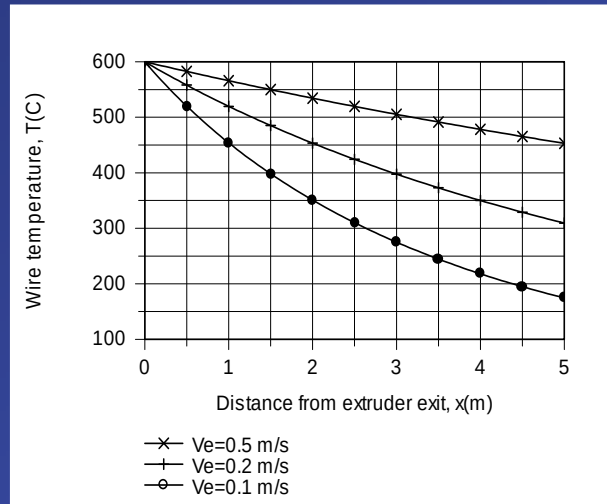
$$T_o = 340^\circ\text{C} \quad <$$

(c) Numerically integrating Equation (1) from $x = 0$ to $x = L = 5.0 \text{ m}$, we obtain

$$T_o = 309^\circ\text{C} \quad <$$

Hence, radiation makes a discernable contribution to cooling the wire.

Parametric conditions reveal the following distributions.



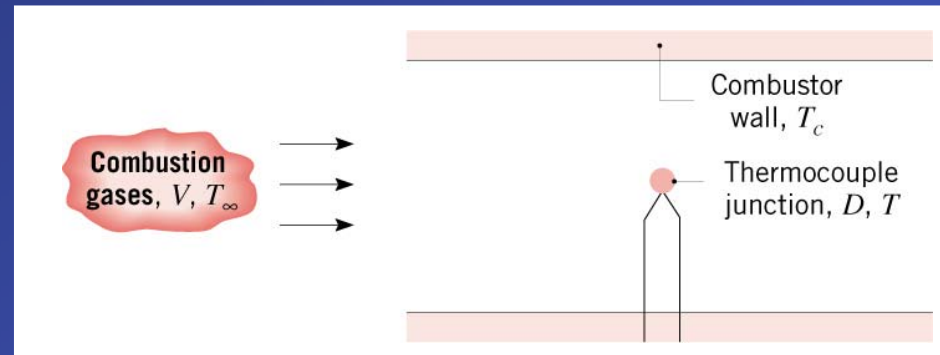
The speed with which the wire is drawn from the extruder has a significant influence on the temperature distribution. The temperature decay decreases with increasing V_e due to the increasing effect of advection on energy transfer in the x direction.

The effect of the surface emissivity is less pronounced, although, as expected, the temperature decay becomes more pronounced with increasing ϵ .

COMMENTS: (1) A critical parameter in wire extrusion processes is the *coiling temperature*, that is, the temperature at which the wire may be safely coiled for subsequent storage or shipment. The larger the production rate (V_e), the longer the cooling distance needed to achieve a desired coiling temperature.

(2) Cooling may be enhanced by increasing the cross-flow velocity, and the specific effect of V may also be explored.

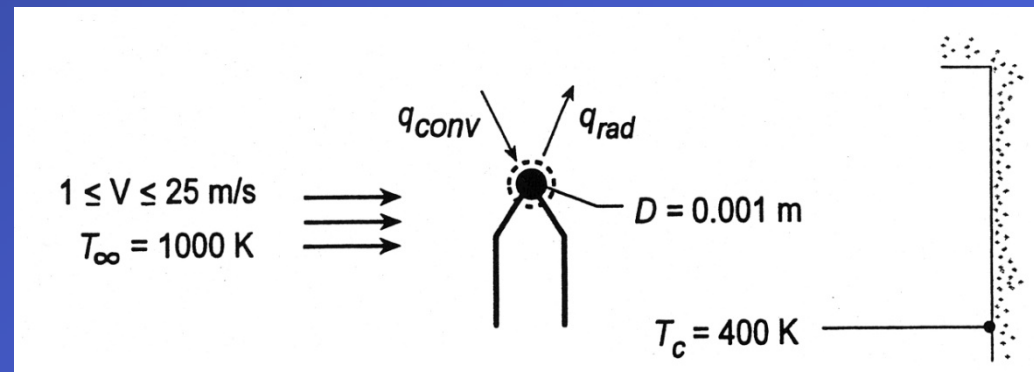
Problem: 7.85 Measurement of combustion gas temperature with a spherical thermocouple junction.



KNOWN: Velocity and temperature of combustion gases. Diameter and emissivity of thermocouple junction. Combustor temperature.

FIND: (a) Time to achieve 98% of maximum thermocouple temperature rise for negligible radiation, (b) Steady-state thermocouple temperature, (c) Effect of gas velocity and thermocouple emissivity on measurement error.

SCHEMATIC:



ASSUMPTIONS: (1) Validity of lumped capacitance analysis, (2) Constant properties, (3) Negligible conduction through lead wires, (4) Radiation exchange between small surface and a large enclosure (parts b and c).

PROPERTIES: Thermocouple: $0.1 \leq \varepsilon \leq 1.0$, $k = 100 \text{ W/m}\cdot\text{K}$, $c = 385 \text{ J/kg}\cdot\text{K}$, $\rho = 8920 \text{ kg/m}^3$; Gases: $k = 0.05 \text{ W/m}\cdot\text{K}$, $\nu = 50 \times 10^{-6} \text{ m}^2/\text{s}$, $Pr = 0.69$.

ANALYSIS: (a) If the lumped capacitance analysis may be used, it follows from Equation 5.5 that

$$t = \frac{\rho V c}{\bar{h} A_s} \ln \frac{T_i - T_\infty}{T - T_\infty} = \frac{D \rho c}{6 \bar{h}} \ln(50).$$

Neglecting the viscosity ratio correlation for variable property effects, use of $V = 5 \text{ m/s}$ with the Whitaker correlation yields

$$\overline{Nu}_D = \left(\bar{h} D / k \right) = 2 + \left(0.4 Re_D^{1/2} + 0.06 Re_D^{2/3} \right) Pr^{0.4} \quad Re_D = \frac{VD}{\nu} = \frac{5 \text{ m/s} (0.001 \text{ m})}{50 \times 10^{-6} \text{ m}^2/\text{s}} = 100$$

$$\bar{h} = \frac{0.05 \text{ W/m}\cdot\text{K}}{0.001 \text{ m}} \left[2 + \left(0.4 (100)^{1/2} + 0.06 (100)^{2/3} \right) (0.69)^{0.4} \right] = 328 \text{ W/m}^2 \cdot \text{K}$$

Since $Bi = \bar{h} (r_o/3) / k = 5.5 \times 10^{-4}$, the lumped capacitance method may be used.

$$t = \frac{0.001 \text{ m} \left(8920 \text{ kg/m}^3 \right) 385 \text{ J/kg}\cdot\text{K}}{6 \times 328 \text{ W/m}^2 \cdot \text{K}} \ln(50) = 6.83 \text{ s} \quad <$$

(b) Performing an energy balance on the junction, $q_{\text{conv}} = q_{\text{rad}}$.

Hence, evaluating radiation exchange from Equation 1.7 and with $\varepsilon = 0.5$,

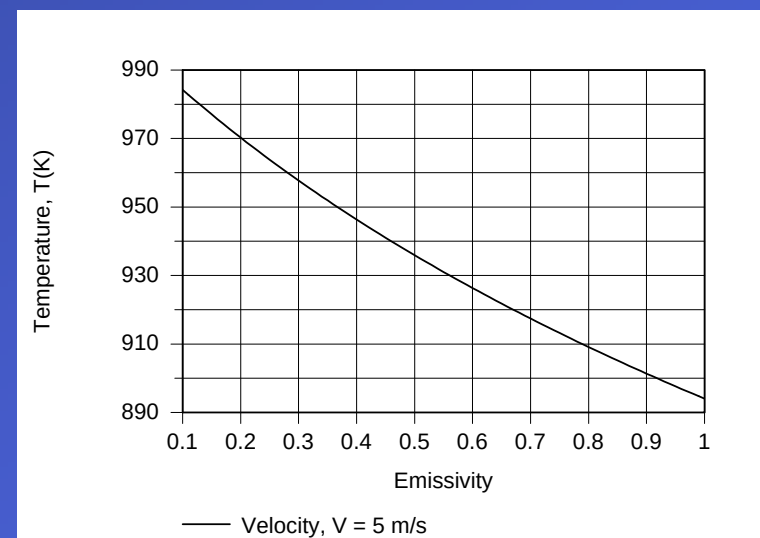
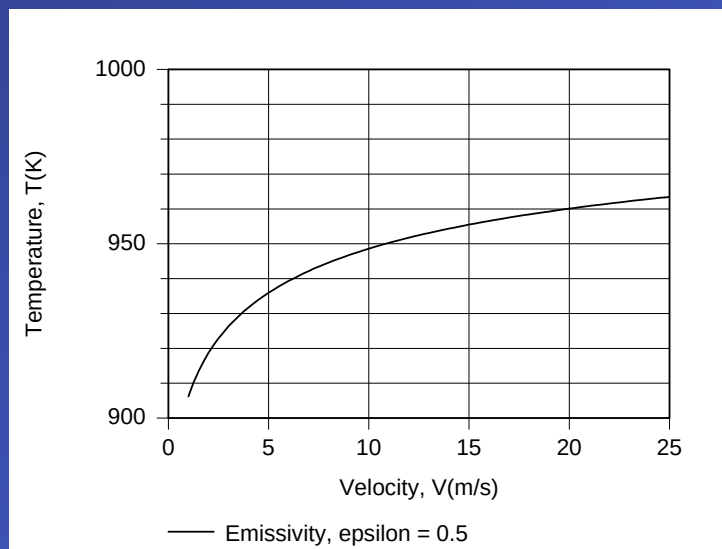
$$\bar{h}A_s(T_\infty - T) = \varepsilon A_s \sigma (T^4 - T_c^4)$$

$$(1000 - T)\text{K} = \frac{0.5 \times 5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4}{328 \text{ W/m}^2 \cdot \text{K}} [T^4 - (400)^4] \text{K}^4$$

$$T = 936 \text{ K}$$

<

Parametric calculations to determine the effects of V and ε yield the following results:



<

Since the temperature recorded by the thermocouple junction increases with increasing V and decreasing ε , the measurement error, $T_\infty - T$, decreases with increasing V and decreasing ε . The error is due to net radiative transfer from the junction (which depresses T) and hence should decrease with decreasing ε .

For a prescribed heat loss, the temperature difference ($T_\infty - T$) decreases with decreasing convection resistance, and hence with increasing $h(V)$.

COMMENTS: To infer the actual gas temperature (1000 K) from the measured result (936 K), correction would have to be made for radiation exchange with the cold surroundings.

What measures may be taken to reduce the error associated with radiation effects?